Introduction to Computational Thinking and Data Science (ICTDS)

Academic Year: 2025/2026

Milestone 1: Climate Stability Analysis: Bragança vs Viseu

Group: Group 11 **Submitted:** March 17, 2026

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1 Introduction

Goal: This report presents a comparative climate stability analysis of two Portuguese cities — Bragança and Viseu — to determine which location is better suited for a **cold-storage logistics facility**. The analysis spans daily meteorological data from 2018 to 2025, evaluating temperature distributions, seasonal variability, extreme weather events, and the thermal load experienced at each site. The key question is: *which city offers a more stable, cooler, and operationally predictable climate?*

2 Dataset Description

2.1 Data Source and Scope

Two CSV datasets were used — `raw_BRAGANCA.csv` and `raw_VISEU.csv` — each containing daily observations. After removing the redundant `time` column and excluding the incomplete year 2026, both datasets span **2018-01-01 to 2025-12-31**, yielding **2,922 daily records per city**.

2.2 Variables

The following features were available or derived from the raw data:

Table 1: Dataset Variables

Variable	Type	Description
`temperature_2m_max`	Numerical	Daily maximum temperature at 2m (°C)
`temperature_2m_min`	Numerical	Daily minimum temperature at 2m (°C)
`temperature_2m_mean`	Numerical	Daily mean temperature at 2m (°C)
`precipitation_sum`	Numerical	Total daily precipitation (mm)
`wind_speed_10m_max`	Numerical	Max wind speed at 10m (km/h)
`wind_gusts_10m_max`	Numerical	Max wind gusts at 10m (km/h)
`shortwave_radiation_sum`	Numerical	Daily solar radiation sum (MJ/m^2)
`sunshine_duration_hrs`	Derived	Sunshine duration (hours; converted from seconds)
`relative_humidity_2m_mean`	Numerical	Mean relative humidity at 2m (%)
`cloudcover_mean`	Numerical	Mean cloud cover (%)
`day_length_hrs`	Derived	Length of day in hours (sunset − sunrise)
`DTR`	Derived	Diurnal Temperature Range = max − min (°C)

2.3 Data Cleaning and Preprocessing

- The redundant `time` column was dropped.
- The `date` column was converted to datetime; `year` and `month` columns were extracted.
- `sunshine_duration` was converted from seconds to hours.
- `day_length_hrs` was derived from sunset and sunrise timestamps; those columns were subsequently dropped.
- Data from year 2026 was excluded as it only covered January–February, which would introduce seasonal bias.
- A `season` categorical variable was created: *Winter* (Dec–Feb), *Spring* (Mar–May), *Summer* (Jun–Aug), *Autumn* (Sep–Nov).

The seasonal approach was chosen over yearly or monthly grouping: yearly grouping conflates opposite temperature regimes, while monthly grouping produces 12 categories — too granular for a summary analysis.

3 Methodology

3.1 Main Questions

- Which city has lower mean temperatures across all seasons?
- Which city experiences fewer extreme heat days (≥25°C, ≥30°C, ≥35°C)?
- Which city has a more stable temperature regime (lower DTR, lower inter-annual variability)?
- Which meteorological features most strongly correlate with temperature and thermal stability?
- How many anomalous temperature days does each city experience per season?

3.2 Analysis Approach

The analysis followed these steps:

1. **Data loading and cleaning** — removal of redundancies, unit conversions, and feature derivation (DTR, day length, sunshine hours).
2. **Seasonal categorization** — assigning each record to one of four seasons and excluding the partial year 2026.
3. **Summary statistics** — computing mean, standard deviation, coefficient of variation (CoV), median, min, max, and IQR for all features, grouped by season, for both cities.
4. **Temperature Stability Index** — constructing a composite index from three normalized components: within-season SD, mean DTR, and inter-annual CoV of seasonal mean temperature.
5. **Pearson correlation analysis** — measuring linear relationships between eight meteorological features and three targets: mean temperature, DTR, and the Stability Index.
6. **Extreme event detection** — using seasonal IQR fences ($Q_1 - 1.5 \cdot \text{IQR}$, $Q_3 + 1.5 \cdot \text{IQR}$) to flag anomalous temperature days.
7. **Thermal envelope analysis** — computing monthly mean max, mean, and min temperatures to visualize the annual temperature profile.
8. **Positive deviation analysis** — measuring deviations of daily max temperature above the monthly mean max, and counting high-temperature threshold exceedances.

4 Results

4.1 Summary Statistics — Temperature

Table 2 presents the seasonal mean temperatures and standard deviations for both cities.

Table 2: Seasonal Mean Daily Temperature (°C) — Bragança vs Viseu (2018–2025)

	Winter		**Spring**		**Summer**		**Autumn**	
City	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Bragança	6.07	2.89	11.28	4.19	21.14	3.92	13.61	4.70
Viseu	8.41	2.70	12.91	4.03	21.49	4.03	15.30	4.55

Bragança is consistently cooler than Viseu in every season. The temperature gap is most pronounced in **Winter** (2.34°C difference) and **Autumn** (1.69°C difference). Summer means are very close (21.14°C vs 21.49°C), while the spread (SD) is comparable across all seasons.

4.2 Diurnal Temperature Range (DTR)

Table 3 shows the seasonal mean DTR for both cities.

Table 3: Seasonal Mean Diurnal Temperature Range — DTR (°C)

City	**Winter**	**Spring**	**Summer**	**Autumn**
Bragança	7.82	10.50	12.95	9.33
Viseu	7.75	9.83	12.87	8.99

DTR values are broadly similar across cities. Bragança shows slightly higher DTR in Spring and Autumn, indicating marginally wider daily temperature swings in those seasons.

4.3 Temperature Stability Index

The Stability Index was computed as the mean of three normalized components: within-season SD, mean DTR, and inter-annual CoV. A higher index indicates greater stability.

Table 4: Temperature Stability Index by Season (0–1 scale; higher = more stable)

City	**Winter**	**Spring**	**Summer**	**Autumn**
Bragança	0.260	0.200	0.238	0.293
Viseu	0.374	0.241	0.216	0.333

Viseu scores higher in Winter and Autumn stability. However, **Bragança scores slightly higher in Summer** (0.238 vs 0.216), which is the most operationally critical season for a cold-storage facility. The indices are modest across both cities, reflecting that Portuguese climate carries meaningful seasonal variability.

4.4 Pearson Correlations with Temperature Targets

Table 5 summarizes the Pearson correlations between the eight meteorological features and mean daily temperature (T_{mean}), DTR, and Stability Index.

Table 5: Pearson Correlations: Features vs Temperature Targets (averaged across both cities)

Feature	$\mathbf{T_{mean}}$	**DTR**	**Stability Index**
Sunshine Duration (hrs)	+0.57	+0.77	−0.44
Shortwave Radiation (MJ/m^2)	+0.73	+0.80	−0.59
Cloud Cover (%)	−0.35	−0.69	+0.12
Relative Humidity (%)	−0.64	−0.78	+0.27
Precipitation (mm)	−0.17	−0.49	+0.10
Wind Speed Max (km/h)	−0.05	−0.31	−0.06
Wind Gusts Max (km/h)	−0.02	−0.31	−0.09
Day Length (hrs)	+0.72	+0.54	−0.72

The strongest positive drivers of temperature and DTR are **solar radiation**, **sunshine duration**, and **day length**. Relative humidity and cloud cover show the strongest negative associations. Day length is particularly notable: it has the highest correlation with Stability Index ($r \approx -0.72$ for Viseu, $r \approx -0.55$ for Bragança), indicating that longer days are associated with lower thermal stability — consistent with higher summer variability.

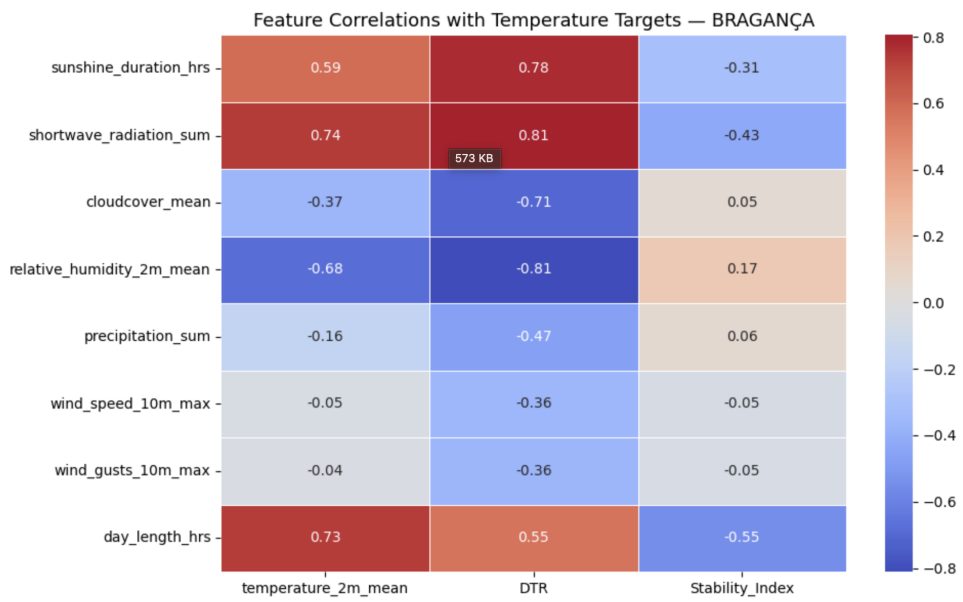

Figure 1: Feature correlation heatmap with temperature targets — Bragança

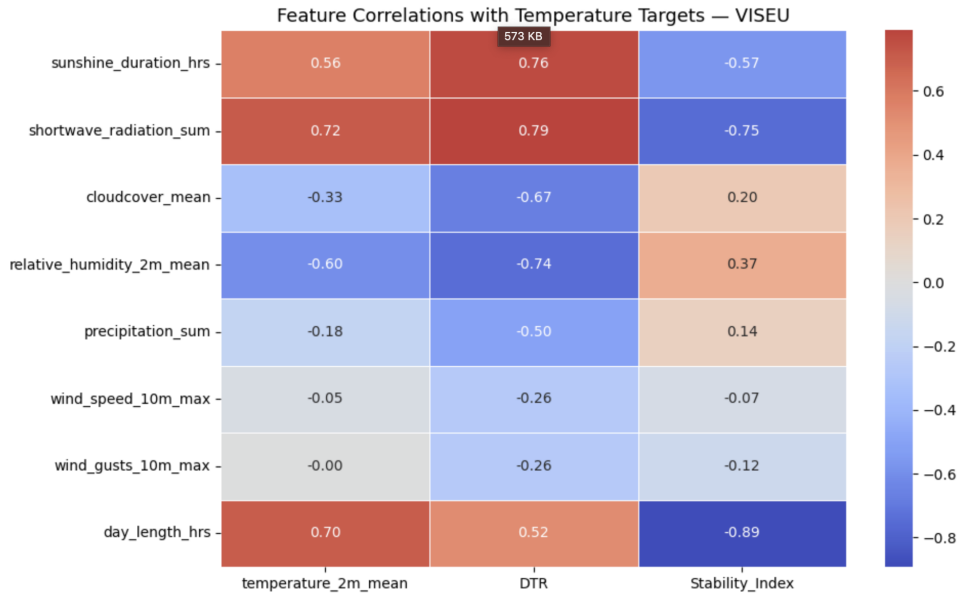

Figure 2: Feature correlation heatmap with temperature targets — Viseu

4.5 Extreme Temperature Events

Seasonal IQR fences were used to detect anomalous days. Results are summarized in Table 6.

Table 6: Extreme Temperature Events by Season

City	Total	Winter	Spring	Summer	Autumn
Bragança	7	0	6	1	0
Viseu	6	0	3	1	2

Bragança has 7 extreme days vs Viseu's 6. The bulk of Bragança's extremes occur in Spring

(6 events), while Viseu distributes across Spring and Autumn. On extreme days, both cities show markedly lower humidity, higher solar radiation, and higher DTR compared to normal days, indicating hot and dry anomalous events.

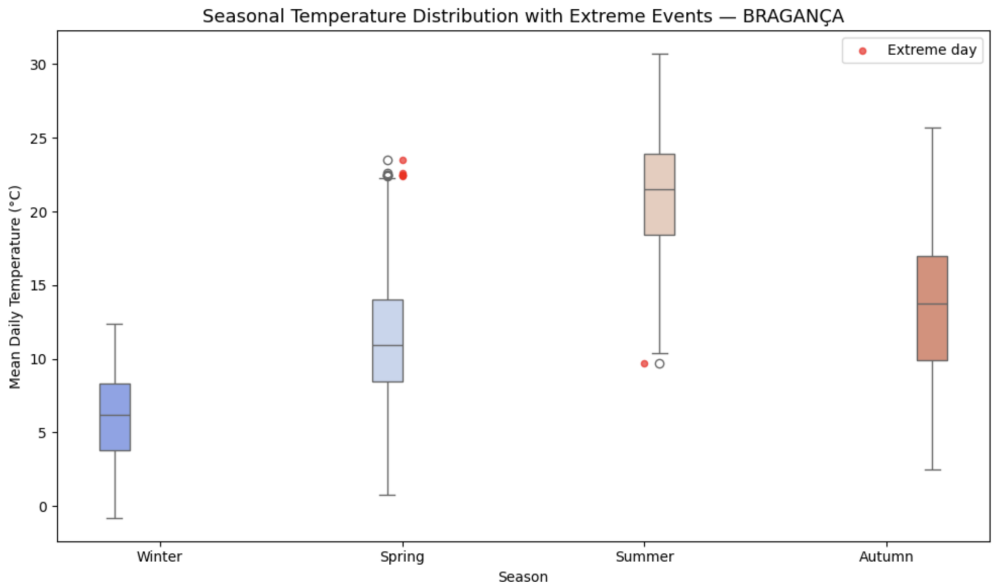

Figure 3: Seasonal temperature distribution with extreme events highlighted — Bragança

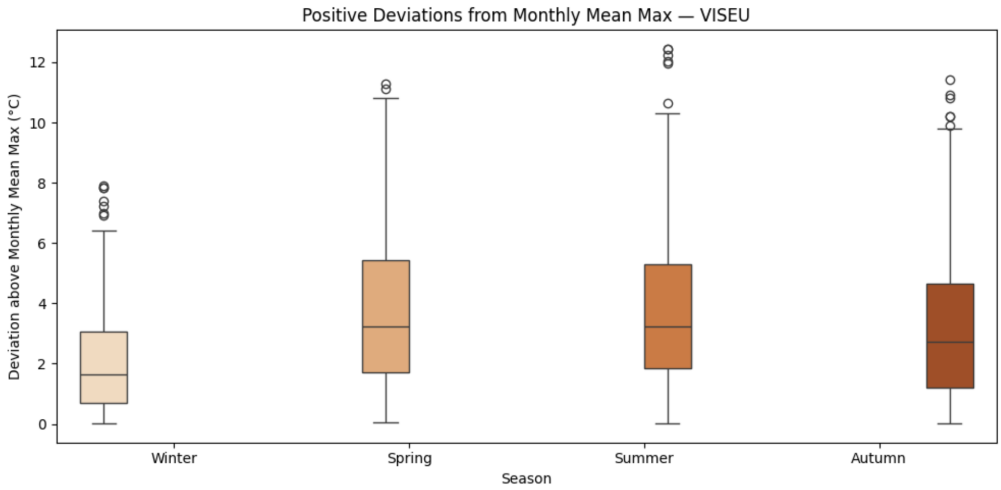

Figure 4: Seasonal temperature distribution with extreme events highlighted — Viseu

4.6 High-Temperature Day Frequency

Table 7: Frequency of High-Temperature Days (2018–2025)

City	**Days ≥ 25°C**	**Days ≥ 30°C**	**Days ≥ 35°C**
Bragança	700 (24.0%)	283 (9.7%)	35 (1.2%)
Viseu	786 (26.9%)	352 (12.0%)	64 (2.2%)

Viseu experiences substantially more high-temperature days at every threshold. The difference is most pronounced at ≥35°C, where Viseu sees **83% more extreme-heat days** than Bragança

(64 vs 35). This represents a significant advantage for Bragança from a refrigeration and energy cost perspective.

4.7 Thermal Envelope

The thermal envelope visualizes the monthly range of temperatures — mean daily max, mean, and min — across the full year. This provides an intuitive view of the annual temperature profile for each city.

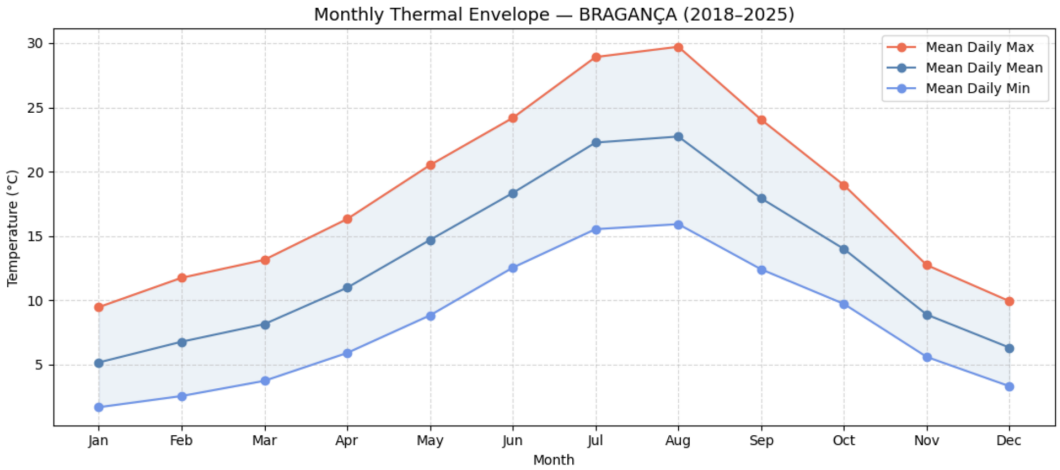

Figure 5: Monthly thermal envelope (2018–2025) — Bragança

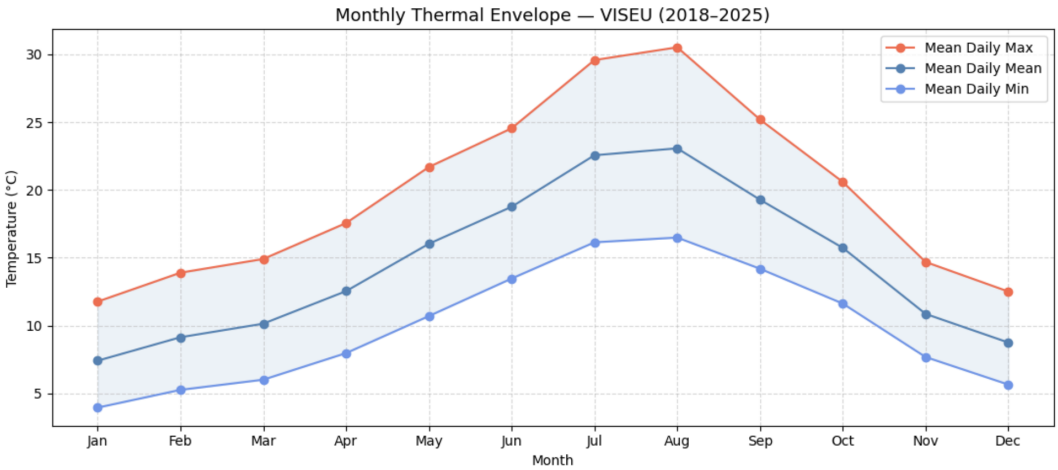

Figure 6: Monthly thermal envelope (2018–2025) — Viseu

4.8 Positive Deviations from Monthly Mean Maximum

The boxplots below show the seasonal distribution of positive deviations — days where the maximum temperature exceeded the monthly mean maximum. These events represent thermal stress periods with elevated cooling demand.

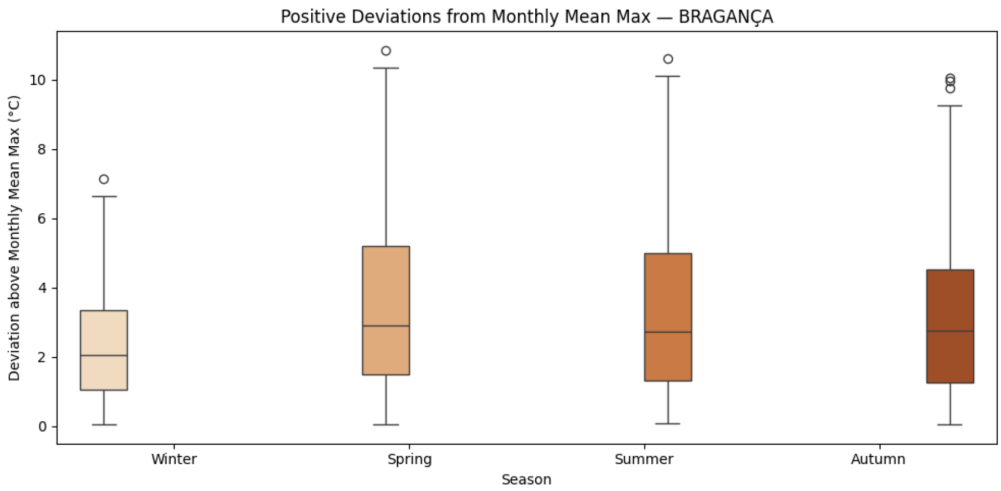

Figure 7: Positive deviations from monthly mean max temperature by season — Bragança

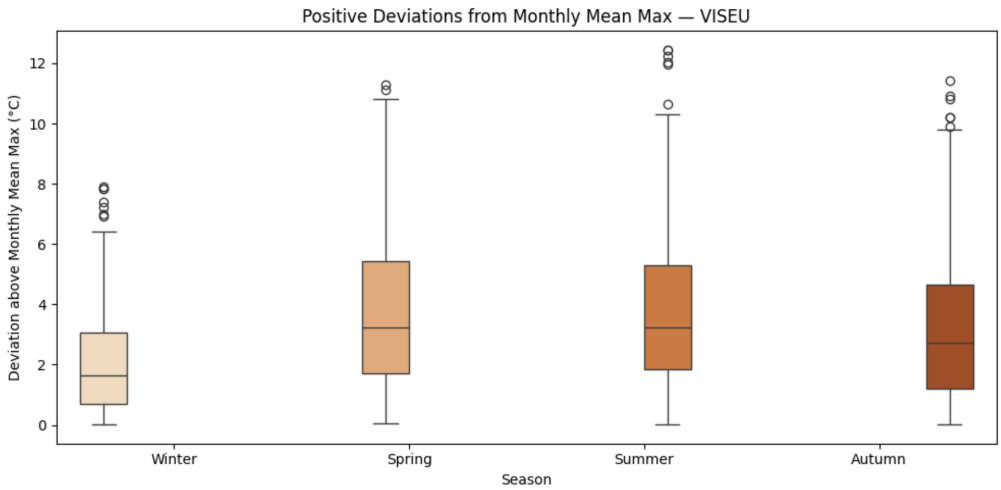

Figure 8: Positive deviations from monthly mean max temperature by season — Viseu

4.9 Summary Comparison Table

Table 8: Head-to-Head Comparison: Bragança vs Viseu

Metric	**Bragança**	**Viseu**
Winter Mean Temp (°C)	6.07	8.41
Summer Mean Temp (°C)	21.14	21.49
Days $\geq$ 25°C	700	786
Days $\geq$ 30°C	283	352
Days $\geq$ 35°C	35	64
Extreme Events (IQR method)	7	6
Summer Stability Index (0–1)	0.238	0.216
Autumn Stability Index (0–1)	0.293	0.333

5 Discussion

The analysis reveals a clear and consistent thermal contrast between the two cities. Bragança is colder across every season, with the largest gap occurring in Winter (2.34°C) and Autumn (1.69°C). While summer mean temperatures are nearly identical, Bragança's advantage becomes stark when considering threshold exceedances: it records 83% fewer days above 35°C than Viseu, a critical consideration for the energy efficiency of a cold-storage facility.

The Stability Index tells a more nuanced story. Viseu shows superior stability in Winter and Autumn, reflecting its more maritime-influenced climate. However, Bragança edges ahead in Summer stability (0.238 vs 0.216), which is operationally significant — Summer is when refrigeration demand peaks.

The correlation analysis confirms that solar radiation and humidity are the dominant drivers of temperature and DTR variability. Day length emerges as the strongest single predictor of the Stability Index for both cities, underlining the seasonal structure underlying the data.

Extreme event detection via IQR fences reveals that both cities are largely free from anomalous cold events. Bragança's 7 extreme days cluster in Spring (6 events), while Viseu's 6 events are spread more evenly. Profiling these days confirms they are primarily hot-and-dry anomalies with elevated solar radiation and reduced humidity — consistent with Iberian heat-wave dynamics.

6 Conclusion

Recommendation: Bragança is the preferred location for the cold-storage logistics facility.

The key drivers of this recommendation are:

- **Lower mean temperatures across all seasons** — particularly in Winter and Autumn, reducing baseline refrigeration load.
- **Significantly fewer extreme heat days** — 83% fewer days ≥35°C and 17% fewer days ≥25°C compared to Viseu.
- **Higher Summer Stability Index** — Bragança's temperature is marginally more predictable during the peak operational season.
- **Lower total thermal load** — fewer cumulative days requiring active cooling above critical thresholds.

Viseu's marginal advantages — slightly superior Winter and Autumn thermal stability, and one fewer IQR-defined extreme event — do not outweigh Bragança's clear edge in overall thermal load and high-temperature frequency. From a logistics and energy cost perspective, Bragança offers the more favorable climate profile.